

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Cross-Text Connections	Hard

Question

Text 1

Ecologists have long wondered how thousands of microscopic phytoplankton species can live together near ocean surfaces competing for the same resources. According to conventional wisdom, one species should emerge after outcompeting the rest. So why do so many species remain? Ecologists’ many efforts to explain this phenomenon still haven’t uncovered a satisfactory explanation.

Text 2

Ecologist Michael Behrenfeld and colleagues have connected phytoplankton’s diversity to their microscopic size. Because these organisms are so tiny, they are spaced relatively far apart from each other in ocean water and, moreover, experience that water as a relatively dense substance. This in turn makes it hard for them to move around and interact with one another. Therefore, says Behrenfeld’s team, direct competition among phytoplankton probably happens much less than previously thought.

Based on the texts, how would Behrenfeld and colleagues (Text 2) most likely respond to the “conventional wisdom” discussed in Text 1?

Answer

- A. By arguing that it is based on a misconception about phytoplankton species competing with one another
- B. By asserting that it fails to recognize that routine replenishment of ocean nutrients prevents competition between phytoplankton species
- C. By suggesting that their own findings help clarify how phytoplankton species are able to compete with larger organisms
- D. By recommending that more ecologists focus their research on how competition among phytoplankton species is increased with water density

Correct Answer: A

Rationale

Choice A is the best answer because based on Text 2, it represents how Behrenfeld and colleagues would most likely respond to the “conventional wisdom” discussed in Text 1. The conventional wisdom cited holds the opinion that when there is species diversity within a phytoplankton population, “one species should emerge after outcompeting the rest”—that is, after being so successful in competing for resources that the other species vanish from the population. However, Text 2 explains that according to Behrenfeld and colleagues, phytoplankton are so small and spaced so far apart in the water that there is “much less” direct competition for resources within phytoplankton populations than scientists had previously thought.

Choice B is incorrect because Text 2 never discusses whether routine replenishment of ocean nutrients affects competition between phytoplankton species. Choice C is incorrect because the interspecies competition discussed in both texts is specifically between phytoplankton species, and neither text considers whether phytoplankton compete for resources with larger nonphytoplankton species. Choice D is incorrect because according to Text 2, Behrenfeld and colleagues argue that water density decreases, not increases, competition between phytoplankton species.